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Opening the Strategic Pandora Box: Conditional Transaction Mechanisms

Yonatan Sompolinsky[1] Ori Newman[2]
Xintong Wang[3] David Parkes[1]

[1]Computer Science, Harvard John A. Paulson School of Engineering and Applied Sciences, Harvard University
[2]Blavatnik School of Computer Science and AI, Tel Aviv University
[3]Department of Computer Science, Rutgers University

Abstract

Conditional transaction engines (CTEs) execute conditional instructions for offline users. This paper formalizes the mechanism-design problem within each engine invocation. A conditional transaction mechanism (CTM) decides which pending conditions to evaluate first because each evaluation delays the eventual write. We model this problem as Strategic Pandora, a discounted variant of the Pandora's box model with independent Bernoulli boxes. Agents report privately assessed success probabilities and, in the full model, values for the write action. We propose the reported-Weitzman mechanism (RW) and the reported-values second-price mechanism (RWSP). To compare revenue without a common prior, we introduce dynamic No-Betting Revenue. Under the stated competition and equilibrium conditions, every qualifying pure equilibrium of RW or RWSP earns a constant fraction of its corresponding dynamic NBR benchmark.

1 Introduction

On October 10, 2025, an estimated $3.21 billion in leveraged positions in crypto was liquidated in a single minute at the peak of a market cascade. Reports at the time put total liquidations at more than $19 billion across roughly 1.6 million trading accounts. In November 2022, FTX customers became unable to withdraw their assets when the exchange halted withdrawals. In a later submission to the CFTC, Better Markets raised the general concern that automatic liquidation in markets that operate around the clock can catch users off guard or asleep as liquidation thresholds approach (Marshall 2025; Axios 2025; CFTC 2022; Better Markets 2025).

Running a 24/7 monitoring server is unsustainable for non-expert traders and family offices. In that, our framework is appealing for the long tail of users who cannot run a 24/7 monitoring server. Moreover, servers can fail during network stress and therefore do not guarantee access during sudden shocks such as the October cascade. The framework is therefore also useful for expert users and entities whose monitoring servers may lose access when the entire system is stressed. The FTX collapse involved a platform-level failure that our framework does not address, but it shows that access to a centralized platform cannot be assumed. These cases motivate protection that can act on behalf of offline users without requiring them to transmit new instructions in real time.

We consider conditional transaction engines (CTEs), which evaluate and sequence instructions while their users are offline. Users submit "If This Then That" (IFTTT) instructions in advance.

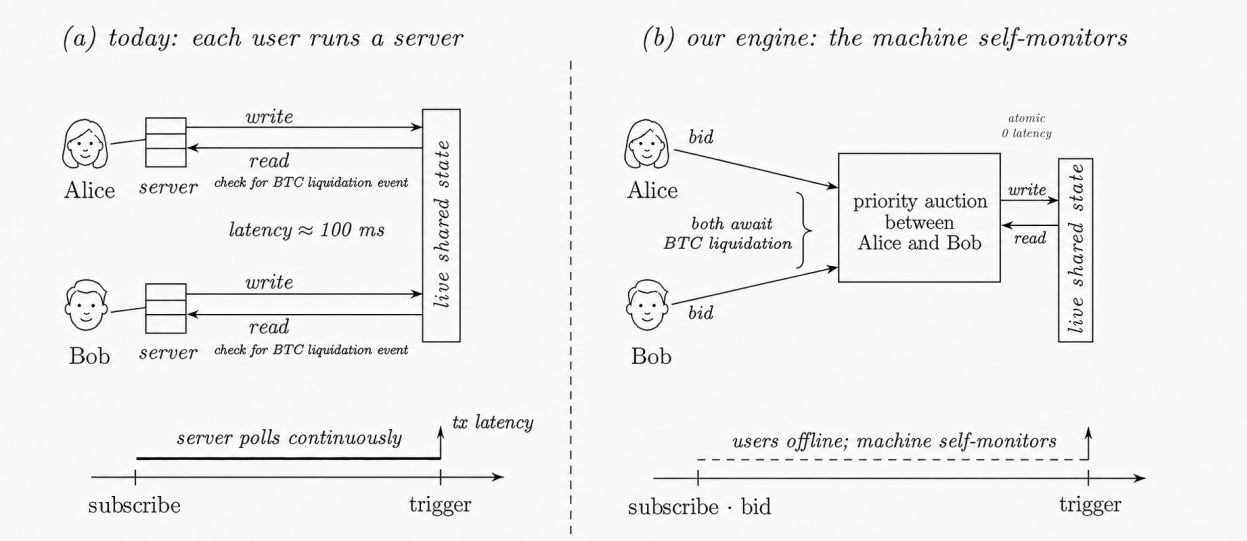


Figure 1: Two ways to act on a BTC liquidation event. (a) Alice and Bob each maintain an online process that reads the shared state and later submits a transaction. (b) Alice and Bob submit persistent value and priority information before the event; the engine evaluates the recorded conditions and selects among the resulting eligible transactions. The "0 latency" label denotes removal of an additional user-side reaction step, not zero physical or network latency. The comparison presumes neither admission nor execution.

At each invocation, the CTE invokes a conditional transaction mechanism (CTM). The CTM uses reported condition probabilities and write values to determine which conditions to evaluate and which instruction receives the next transaction slot. This architecture extends the capability of existing transaction engines, such as Ethereum's EVM, which sequence transactions in real time only and cannot trigger transactions autonomously. CTEs remain relevant even when the system is run by one logical server or sequencer, as at a centralized exchange such as Nasdaq. However, as we explain, the primary use case is expressive composable systems such as DeFi, where transactions can interact atomically with multiple markets or logic zones. Composite transactions render Price First Time Second and similar TradFi policies unsuitable, since no single price can rank a composite rule. Submission time too is irrelevant, because rules may be submitted days ahead of their triggering.

An empirical study of Aave liquidations on October 10 documents speculative traders repeatedly submitting transactions in anticipation of liquidations. One Aave liquidator submitted 14,333 transactions in an attempt to time the event, only to have 99.05% of those transactions reverted (Sevim and Ferreira Torres 2026, Section 6.2 and Table 9). Our framework obviates such laborious submission processes: The user submits one persistent condition or action rule, and the rule remains available to the sequencer across invocations. The sequencer evaluates its condition and, when the condition is satisfied and the rule's user is selected in the auction, enacts on the user's behalf with theoretically zero latency. Furthermore, when the condition holds but some other participant outbids the user, the mechanism avoids charging the latter, relieving the costs of participation.

In fact, the very need for liquidators stems from smart contracts' inability to trigger their own conditionals. Lacking such capabilities, DeFi lending contracts incentivize and rely on third parties to listen to on-chain state (e.g., an oracle update on Chainlink contract) and trigger liquidation when needed (Aave n.d.). Our paradigm alleviates such workarounds: Using a CTE, the maintainers can encode in advance the entire liquidation logic as a conditional instruction, and avoid manual

liquidators altogether.[1]

1.1 Existing approaches

Delegated execution is not new. Brokers in TradFi and keepers and solvers in DeFi monitor specified conditions and submit trades or transactions when those conditions hold (Gelato Network 2023; Chainlink 2023; Anoma 2024). More generally, subscription and listener interfaces, event-driven programming systems, IFTTT-style automation, and platforms such as Reactive Network evaluate predicates and dispatch actions (Kaleem et al. 2021; Zhao et al. 2022; Reactive Network 2024). However, these services operate outside the sequencing engine that commits the final state transition. They can attempt submission, but they do not determine admission or priority. The mechanism-design problem therefore remains: how should the sequencing engine allocate condition reads and the next transaction slot among pending rules?

Event-based smart contract systems were proposed in the past, albeit absent a mechanism-design-informed architecture. These systems largely treat triggered transactions as non-mandatory inputs for the sequencer, and do not treat evaluation inside their systems. This renders those systems an incremental improvement over existing smart contract systems. In contrast, a CTE can define deterministic mandatory prioritization that operates in-consensus, regardless of the operator's cooperation. Its CTM treats reads and writes in the same mechanism.

1.2 The one-shot allocation problem

Evaluating a rule's condition and altering the shared state constitute two distinct resource-allocation problems. The former is a 'read', and as such it is commutative: it consumes computation, but does not inherently preclude evaluation of other rules, and the ordering has no consequence. Executing a satisfied rule, in contrast, alters the shared state and does not generally commute with other writes. The mechanism allocates at most one rule to the next transaction slot.

Evaluating too many conditions still imposes a prohibitive externality on the system, since each read delays further the eventual write. We model the process through a discount factor applied to the value of the write.

Coupling thus the reads and writes allows for more efficient mechanisms that avoid spending read cycles if the corresponding write is outbid by bidders on other queries. For instance, conditional instructions of the Aave liquidator discussed above will not be read and evaluated if the liquidator is not guaranteed to receive priority on the write, as it is still outbid by counterparts.

The resulting problem maps into a discounted variant of the Pandora's box problem with independent Bernoulli boxes, where each pending rule is represented by a box with unknown parameters. Crucially, the system is dynamic: rules' success probabilities may change with the state and environment, and agents may continuously revise their bids accordingly. Our paper focuses on the allocation of reads and writes at a given point in time. We leave the broader dynamic process for separate work.

In our multi-agent version of Pandora, agents report the `True` probability of pending rules they are interested in. Agents may hold very different beliefs, and the mechanism remains prior-free and does not aim to learn the real underlying parameter. [2]

[1] For liquidation to be applied atomically with its triggering condition, the maintainors should attach sufficient bids. In our paradigm, they can attach bids for 'write' only, and rely on the reports of online 'read' agents to increase the 'read' bids according to real-time pressure.

[2] This separates our framework from traditional exploration-exploitation problems, and from no-regret framework. This is motivated by the environment being noisy and dynamically changing, and by one primary use case being protection from market catastrophes which is inherently not learnable.

In the full *reported values* model, agents report also their value or bid for the write action. However, for pedgagoical reasons, we study also an interim model, the *known box values*, where box values are fixed and known, and agents report read-probabilities only.

For the one-shot game, we introduce two mechanisms, one for each model: the reported-Weitzman mechanism (RW) and the reported-values second-price mechanism (RWSP). Both mechanisms adapt Weitzman's optimal solution to the planner's Pandora problem to a mechanism-design setting in which the mechanism must choose a payment vector. They differ in how payments are discounted and in whether the write action is allocated as in a first-price or second-price auction. Section 2.3 formalizes both mechanisms and explains these differences. We prove that each mechanism obtains a constant fraction of the optimal mechanism in the corresponding model. The benchmark restricts outcomes to an admissibility class which we now discuss.

1.3 Prior-free mechanism design

Mechanism design typically strives to maximize social welfare. However, when agents hold different beliefs about the underlying uncertainty, and the mechanism itself is restricted to remain prior-free, social welfare is undefined. Summing agents' subjective utilities leads to betting, mnamely, to allocations that count ex-ante value from transfers between agents holding opposing beliefs on the likelihood of events, rather than from exogenous utility.

Drawing inspiration from Gilboa et al.'s No-Betting-Pareto Dominance (NBPD) criterion, we introduce No-Betting Revenue (NBR) and dynamic NBR (dNBR) to define an admissibility class of outcomes. These criteria ensure that the framework does not reward mechanisms that obtain an artificially high revenue due to betting. We formalize the admissibility class in Section 3, and show why the NBPD criterion is inadequate for a mechanism design setup.

1.4 Our contributions

- We define the CTM, the one-invocation framework for conditional delegated execution. A CTM takes the active conditional rules, reports, and bids; determines which conditions to evaluate and in what order; assigns payments; and selects the next write. We show how this framework maps into the Strategic Pandora game.
- We introduce a new prior-free criterion for mechanism design with heterogeneous beliefs: NBR for a single decision and dynamic NBR for sequential decisions. NBR includes the auctioneer in the comparison and ranks admitted outcomes by deterministic revenue. Dynamic NBR uses range-constrained product beliefs to account for the value generated at sequential decisions. This contribution may be of independent interest.
- We devise the RW mechanism for known box values and the RWSP mechanism for reported values. Under the stated competition and equilibrium conditions, every qualifying pure equilibrium of each mechanism earns a constant fraction of the corresponding dynamic NBR revenue benchmark.

1.5 Conditional Transaction Engines

The transaction seletion mecahnism, the CTM, is only one component within the broader engine, the CTE. The former is responsible, at the end of every transction slot, to process a certain snapshop of conditional rules and their bids, and output the active rule(s) to evaluate, and to allocate the next transaction slot.

The CTE handles the dynamic process of running a series of CTM instances, one after each transaction slot. The CTE should ideally comprise other protocols too, in particular ones governing

bid submisson and revision. A properly devised CTE protocol would ensure that the operator cannot engage in bid sniping or free-ride the probability-agents' information.

A hollistic discussion of the CTE, and of the operator's restricted role in real-time admission, will appear in a separate work. For our scope, one can safely assume away the opeartor altogether, and focus on the question of choosing reads and write based on a fixed input of conditional rules and their bids. The reader should further think of real-time transactions as special trivial conditional rules of the form If `True` Then That.

2 Strategic Pandora and RW

After each trasaction slot executes, the CTM chooses the next rules (or regular If `True` Then That tranasction) to evaluate and own the next write slot. It does so based on a fixed set of read (probaboltiy) and write (value) bids. As mentioned above, the problem of the CTM designer can be formalized as a prior-free variant of the Pandora box problem.

2.1 Discounted Pandora with Bernoulli boxes

There are M independent boxes. Box j contains $v_j > 0$ with probability p_j and zero otherwise. Inspection is free but creates delay: a common $0 < \delta < 1$ discounts the box in position t by δ^{t-1}, and search ends at the first success. For an order $\pi = (\pi_1, \dots, \pi_M)$, its value is

$$W_\delta(p, v; \pi) = \sum_{t=1}^{M} \delta^{t-1} \Big(\prod_{s<t} (1 - p_{\pi_s}) \Big) p_{\pi_t} v_{\pi_t}, \qquad W_\delta(p, v) = \max_{\pi} W_\delta(p, v; \pi).$$

The discounted Weitzman index is $z(p, v) = pv/(1 - \delta + \delta p)$; box j has index $z_j = z(p_j, v_j)$.

The following ordering lemma determines the nonstrategic benchmark used throughout.

Lemma 1 (Discounted Weitzman optimality). *Among policies that execute the first successful box and then stop, ordering the boxes in weakly decreasing order of z_j maximizes W_δ.*

Appendix A proves the lemma by comparing adjacent boxes and repeatedly swapping inverted pairs.

2.2 Agents and reports

There are n agents. Agent i privately assigns box j success probability $p_i^j \in (0, 1)$, independently across boxes.[3] Under known box values, v_j is public and the agent reports $q_i^j \in [0, 1]$. Under reported values, its true type is (p_i^j, v_i^j) and it reports (q_i^j, b_i^j), with $b_i^j \geq 0$. For every box, write $\widehat{q}_j = \max_i q_i^j$ and let $r(j)$ be its selected highest probability reporter. In RW under known box values, the selected reader also receives the box's value if that box is the first success. Under reported values, reader and writer roles are selected separately as specified below.

[3]The restriction to interior true beliefs is imposed to exclude boundary cases immaterial to the main comparison results.

2.3 The two mechanisms

Mechanism 1: RW under known box values

Input: Public values v_j; reports q_i^j.
Output: An inspection order, up-front payments, and one executed box.

1. **Adopt reports.** Set $\widehat{q}_j = \max_i q_i^j$ and select a reader $r(j)$ attaining the maximum.
2. **Rank.** Set $\widehat{z}_j = \widehat{q}_j v_j/(1-\delta+\delta\widehat{q}_j)$ and sort decreasingly.
3. **Charge.** At position t, set $D_j = \delta^{t-1}\prod_{s<t}(1-\widehat{q}_{\pi_s})$ and charge the reader $P_j = D_j\widehat{q}_j v_j$ up front.
4. **Run.** Inspect in order and execute the first success.

Mechanism 2: RWSP

Input: Reports (q_i^j, b_i^j).
Output: An inspection order, reader and writer roles, payments, and one executed box.

1. **Select roles.** Set $\widehat{q}_j = \max_i q_i^j$, reader $r(j)$ at that maximum, writer $w(j)$ at the highest value report, and second price $y_j = b_j^{(2)}$.
2. **Rank.** Set $\widehat{z}_j = \widehat{q}_j y_j/(1-\delta+\delta\widehat{q}_j)$ and sort decreasingly.
3. **Charge.** Set $D_{r(j),j} = \delta^{t(j)-1}\prod_{k\prec j}(1-q_{r(j)}^k)$ and charge the reader $D_{r(j),j}q_{r(j)}^j y_j$ up front.
4. **Run.** Inspect in order. At the first success, execute the box and transfer y_j from writer to reader.

In RWSP, $b_j^{(2)}$ is the second-highest value report among distinct agents, with ties occupying separate ranks. A value tie is broken by probability report, a probability tie by value report, and remaining exact ties by a fixed deterministic rule. RW exact ties are also resolved deterministically. The technical equilibrium-selection conditions used by the known-values results are stated in Appendices B and C.

The up-front objective-term payment rule. RW replaces the probability inputs to the discounted Weitzman procedure with the most optimistic reports and attaches payments to the resulting allocation rule. For each box, the selected highest probability reporter pays up front the box's discounted contribution to the Weitzman objective. Raising a probability report can win the reader role and move the box earlier, but it also raises the payment. Lowering a report reduces the payment but risks losing the role to a rival. Because the entire objective term is charged before inspection, realized successes and failures do not change mechanism revenue. Revenue equals the maximum discounted value under the most optimistic report for each box:

$$\mathrm{Rev}_{\mathrm{RW}}(q) = \sum_j P_j = W_\delta(\widehat{q}, v).$$

A two-box instance: order, delay, and payment. The following calculation applies RW to two portfolio-protection rules. It shows why the highest success probability need not be inspected first, and how the same reports determine what each selected reader pays before inspection. Take public write values $v_A = 10$, $v_B = 6$, discount $\delta = 0.9$, and highest probability reports $\widehat{q}_A = 0.5$, $\widehat{q}_B = 0.8$. Substituting into RW gives:

	Rule A	Rule B
Public value v_j	10	6
Adopted report $\widehat{q}_j$	0.5	0.8
Index $z(\widehat{q}_j, v_j)$	$5/0.55 \approx 9.09$	$4.8/0.82 \approx 5.85$
Inspection position	First	Second, if A fails
Discount times reported reach D_j	1	$0.9(1-0.5) = 0.45$
Up-front payment $D_j\widehat{q}_j v_j$	5	$0.45(0.8)(6) = 2.16$

RW inspects A first because its index is larger. If A succeeds, it receives the next transaction slot and B is never inspected. If A fails, B is inspected one period later, so its successful write has discounted value $0.9 \cdot 6 = 5.4$. Both payments have already been collected, including the payment for B when it is never reached; retained revenue is $5 + 2.16 = 7.16$ in every outcome. The calculation applies to the stated reports, without assuming equilibrium.

Separating probability reports from value reports. A direct first-price extension would combine the reader's probability report with the writer's value report, charge the reader the box's objective term, and charge the writer its own report after success. Pure equilibrium may then permit writer overbidding, so the success transfer can exceed the selected writer's true value and the dynamic NBR membership argument fails. RWSP gives execution upon success to the writer with the highest value report and computes the box's index using the second-highest reported value. This second-highest price also determines the writer's payment to the reader upon a successful read. In the value-only, single-box specialization stated below, truthful value reporting is weakly dominant. The discount on the reader's up-front payment is calculated using that agent's reports about the previous boxes, rather than the adopted maximum reports. An ex-ante charge to the writer is not a substitute: the writer may assign very low probability to the condition while placing very high value on priority if it occurs.[4]

Remark (second-price truthfulness and offline value submission). Consider a user protecting one financial position in a particular contract or liquidity pool. For simplicity, assume that the user is single-minded: it has positive value for the corresponding protection rule and zero value for all other boxes or events. Assume also that it does not compete for the reader role. Holding the probability side and the other value reports fixed, writer allocation and payment for this box are those of a standard second-price auction. Truthful value reporting is therefore weakly dominant. The value player may be a trader or high-net-worth investor rather than a server operator. It reports only how much execution is worth to it; it need not estimate or monitor the probability that the condition holds. In the application, the division is temporal as well as informational: probability players are typically online and supply current probability reports, whereas the value player may have less information about real-time market conditions and remain offline. Probability–value decoupling therefore permits the value player to benefit from the information supplied by these probability players. It can submit its value and its execution and payment authorizations in advance without operating a server continuously or communicating a fresh instruction during the event. Once these inputs are admitted to the mechanism state, going offline or losing real-time access creates no additional loss of allocation or utility by itself.

Applied to October 10, this specialization removes two requirements from portfolio protection: the value owner need not remain awake, and a previously admitted instruction need not be transmitted when its condition holds. The statement concerns participation by an already submitted rule, not price formation. It is conditional on automatic execution of the recorded mechanism logic and does not guarantee admission, platform liveness, execution, or that a stored value remains current.

The next section develops NBR for one decision and then dynamic NBR for sequential decisions. Section 4 then gives the membership and revenue guarantees under known box values and reported values.

[4]For example, a large liquidity provider may believe that the market is stable but place a high conditional value on priority in a black-swan crash.

3 No-Betting Pareto Dominance, No-Betting Revenue, and Dynamic No-Betting Revenue

The mechanisms in Section 2 collect up-front payments from agents who may disagree about success probabilities. This disagreement also affects the revenue benchmark. Contingent transfers can cover each agent's up-front payment in expectation under that agent's own belief, even when the mechanism allocates no value. Scaling these transfers then produces arbitrarily large revenue. Individual rationality alone therefore gives no meaningful revenue comparison for RW or RWSP.

The example below isolates the problem for a single decision. NBR excludes outcomes whose revenue is supported only by incompatible beliefs. In Strategic Pandora, agents may also disagree about the probability that a later box is reached. Dynamic NBR applies the same restriction to the sequence of inspections.

3.1 Example: revenue without allocated value

The decisions and transfers. There are two agents and two independent binary conditions A, B. The mechanism observes A first and observes B only if A fails; it stops at the first success. The discount is $\delta = 1/2$. Before either observation, each agent pays $\Lambda > 0$ to the mechanism. Subsequent transfers are between the agents, as follows:

Observed event	Transfer between agents	Discount
A succeeds	Agent 2 pays $20\Lambda/13$ to agent 1	1
A fails, B succeeds	Agent 1 pays $64\Lambda/13$ to agent 2	1/2
Both fail	No transfer	—

The mechanism retains the two up-front payments in every outcome. The two agents have opposed interior beliefs:

$$(p_1^A, p_1^B) = (3/4, 1/4), \qquad (p_2^A, p_2^B) = (1/4, 3/4).$$

Own-belief individual rationality. Each agent evaluates the discounted amount it receives minus the discounted amount it pays to the other agent. The probability of the second transfer is the probability that A fails times the probability that B succeeds. Under their respective beliefs, these expectations are

$$\text{Agent 1:} \quad \frac{3}{4}\frac{20\Lambda}{13} - \frac{1}{2}\frac{1}{4}\frac{1}{4}\frac{64\Lambda}{13} = \Lambda,$$
$$\text{Agent 2:} \quad -\frac{1}{4}\frac{20\Lambda}{13} + \frac{1}{2}\frac{3}{4}\frac{3}{4}\frac{64\Lambda}{13} = \Lambda.$$

Subtracting each agent's up-front payment Λ gives zero expected utility. Thus own-belief individual rationality permits retained revenue 2Λ for every $\Lambda > 0$, although no value is allocated.

Excluding revenue without allocated value. Each agent regards the transfer it receives as sufficiently likely to cover its up-front payment. Under any one belief used to evaluate both agents, however, their transfers sum to zero at each decision. The revenue comparison must therefore require support from allocated value in addition to own-belief individual rationality.

3.2 NBR: revenue comparison for a single decision

In "No-Betting-Pareto Dominance," Gilboa, Samuelson, and Schmeidler (2014, Definitions 2–3) require every affected agent to gain strictly, both under its own belief and under a single hypothetical belief. We adapt their comparison to mechanism revenue relative to no action.

Include the auctioneer and allow zero agent surplus. Competition can make each agent's payment equal its expected gross payoff under the belief used for the comparison, leaving every agent with zero expected surplus. A revenue benchmark should admit such outcomes. We therefore include the auctioneer in the comparison with no action. Positive revenue can supply the required strict gain while every agent is weakly better off.

Fix one decision with finite uncertain outcome X. Agent i has belief P_i, gross payoff $U_i(X)$, and up-front payment t_i. Gross payoff includes any contingent transfers between agents, which balance in every state. Write $u_i(X) = U_i(X) - t_i$. Let the auctioneer be participant 0, with

$$u_0(X) = R = \sum_{i=1}^{n} t_i.$$

Revenue is deterministic: for the mechanism outcome being evaluated, its amount does not vary with the realized state. The auctioneer supplies no additional belief or strategic report. Normalize every participant's payoff under no action to zero, and retain weak own-belief individual rationality, $\mathbb{E}_{P_i}[u_i] \geq 0$ for each agent i.

The comparison requires one certifying belief ν such that

$$\mathbb{E}_\nu[u_i] \geq 0 \quad (i = 0, \ldots, n), \qquad \mathbb{E}_\nu[u_j] > 0 \quad \text{for some } j \in \{0, \ldots, n\}. \tag{NO-ACTION}$$

Because $u_0 = R$ is deterministic, $R > 0$ satisfies the strict-gain requirement for the auctioneer.

Which certifying beliefs? We require the certifying belief to belong to

$$\mathcal{C} := \operatorname{co}\{P_1, \ldots, P_n\} = \left\{ \sum_{i=1}^{n} \lambda_i P_i : \lambda_i \geq 0,\ \sum_{i=1}^{n} \lambda_i = 1 \right\}. \tag{HULL}$$

This is a further departure from Gilboa, Samuelson, and Schmeidler (2014, Section 4.3), who allow any hypothetical belief. Brunnermeier, Simsek, and Xiong (2014) use the convex hull of agents' beliefs for belief-neutral welfare comparisons. Our certificate is existential; their efficiency and inefficiency classifications quantify over every belief in that set. Here the choice is motivated by sequential mechanism design: we begin with the one-box interval below and later permit a separate product certificate at each decision, with every marginal in its corresponding range. Certification cannot then use probabilities outside those ranges to enlarge the revenue benchmark.

From individual inequalities to a revenue metric. The individual part of (NO-ACTION) is

$$\mathbb{E}_\nu[U_i] \geq t_i \quad (i = 1, \ldots, n), \qquad R \geq 0.$$

Summing the agents' inequalities gives the aggregate certificate

$$0 \leq R \leq \max_{\nu \in \mathcal{C}} \mathbb{E}_\nu \left[\sum_{i=1}^{n} U_i \right]. \tag{NBR-CERT}$$

An outcome is **NBR-admissible** if revenue is deterministic, every agent satisfies weak own-belief individual rationality, and (NBR-CERT) holds. The individual inequalities motivate this definition and are a sufficient way to verify it. The aggregate condition is sufficient for the revenue bound needed by the benchmark: transfers between agents cancel before evaluation, leaving aggregate allocated value. It need not make every agent individually rational under one common belief and does not certify an absence of betting. Strict improvement is not an additional requirement of the aggregate definition, which also admits zero-revenue outcomes, including no action.

Complete ranking within the admissible class. For admissible outcomes a, b, NBR ranks a at least as high as b exactly when $R(a) \geq R(b)$. Deterministic auctioneer revenue therefore gives a complete ranking, with equal revenues receiving equal scores. The same scalar ranks outcomes satisfying dynamic NBR below. For an equilibrium benchmark, take the supremum of R over the specified mechanisms and their admissible equilibrium outcomes.

3.3 One box: probability player, value player, and auctioneer

Let $X \in \{0, 1\}$ indicate success of the single box, put $p_i = P_i(X = 1)$, and write $\underline{p} = \min_i p_i$ and $\overline{p} = \max_i p_i$. Then

$$\mathcal{C} = \{\mathrm{Ber}(\theta) : \theta \in [\underline{p}, \overline{p}]\}.$$

This is a single allocation decision with uncertain execution; there is no later box whose reach must be evaluated.

Suppose the probability player r and value player w are distinct. The value player receives value $v > 0$ upon success and then pays y to the probability player. They pay t_r, t_w up front, and all other agents have zero payoff. Thus

$$U_r(X) = Xy, \qquad U_w(X) = X(v - y), \qquad u_0 = R = t_r + t_w.$$

Their own-belief individual rationality conditions are

$$p_r y - t_r \geq 0, \qquad p_w(v - y) - t_w \geq 0. \tag{ONE-IR}$$

The detailed comparison with no action requires one $\theta \in [\underline{p}, \overline{p}]$ such that

$$\theta y - t_r \geq 0, \qquad \theta(v - y) - t_w \geq 0, \qquad R \geq 0, \tag{ONE-CERT}$$

with at least one inequality strict, including the possibility $R > 0$. NBR uses (ONE-IR) and deterministic revenue together with only the summed requirement

$$0 \leq t_r + t_w \leq \overline{p}\, v.$$

In RWSP, $t_w = 0$. When $0 \leq y \leq v$, (ONE-IR) also implies both agents' individual weak inequalities at $\theta = \overline{p}$. For example, $p_r = p_w = p \in (0, 1)$, $y = v$, $t_r = pv$, and $t_w = 0$ give both agents zero surplus under their own and certifying belief, while the auctioneer gains $pv > 0$. Neither agent needs a strict gain.

3.4 Dynamic NBR: The Sequential Extension of NBR

Sequential decisions create a reach problem absent from NBR. A high probability of early success can justify an early payment but reduces the probability of reaching a later allocation. The later agent may expect early failure and therefore be willing to pay more for that allocation. A belief

that justifies the early payment may consequently fail to justify the later payment. Appendix B gives a two-decision example and the full calculation.

Fix an outcome of a finite-horizon mechanism. Decision t contributes random payoff $U_{i,t}$ to agent i, including discount and the indicator that the decision is reached. Contingent transfers between agents balance at each decision in every state, and agent i pays t_i up front. The dynamic NBR criterion extends NBR in two respects. First, let $\mathcal{P}$ contain the product priors whose marginal probabilities lie between the agents' corresponding marginal beliefs:

$$\mathcal{P} = \left\{ \bigotimes_k \nu_k : \min_i P_i(X_k = x) \le \nu_k(x) \le \max_i P_i(X_k = x) \text{ for every } k, x \right\}. \tag{RANGE}$$

Second, the individual extension of the motivating certificate chooses $\nu^t \in \mathcal{P}$ for each decision t so that

$$\sum_t \mathbb{E}_{\nu^t}[U_{i,t}] - t_i \ge 0 \qquad \text{for every agent } i.$$

Different decisions may use different priors, but within each decision one prior evaluates every agent's payoff and the event of reaching that decision. Summing these individual inequalities motivates the aggregate condition below.

A mechanism outcome satisfies **dynamic NBR** when three requirements hold. First, retained revenue is deterministic and equals $R = \sum_i t_i \ge 0$. Second, every agent has own-belief individual rationality, $\sum_t \mathbb{E}_i[U_{i,t}] \ge t_i$. Third, revenue has the decision-by-decision certificate

$$R \le \sum_t \sup_{\nu \in \mathcal{P}} \mathbb{E}_\nu \left[\sum_i U_{i,t} \right]. \tag{CERT}$$

Deterministic auctioneer revenue ranks the admissible outcomes, and own-belief individual rationality preserves voluntary participation. The aggregate certificate is sufficient to bound admissible revenue by certifiable allocated value; this is the property used by the benchmark proofs. For a fixed allocation and decision structure, adding transfers that sum to zero at each decision in every state does not increase the right-hand side of (CERT). Such transfers can still affect own-belief participation and therefore which outcomes qualify. The dynamic NBR definition does not require an absence of betting or individual rationality under the certifying priors. The membership proofs establish the stronger individual certificate, as recorded in Appendix B.

For one Bernoulli box, there is one decision and no later-box reach probability to evaluate. In this case,

$$\mathcal{P} = \mathcal{C} = \{\mathrm{Ber}(\theta) : \theta \in [\underline{p}, \overline{p}]\},$$

so the sum over decisions in (CERT) has one term and (CERT), together with $R \ge 0$, becomes

$$0 \le R \le \max_{\nu \in \mathcal{C}} \mathbb{E}_\nu \left[\sum_i U_i \right],$$

which is (NBR-CERT). Deterministic revenue and own-belief individual rationality are the same requirements in both definitions. Thus an outcome satisfies dynamic NBR if and only if it is NBR-admissible in the one-Bernoulli-box setting. This is the precise sense in which dynamic NBR extends NBR. With several boxes, by contrast, $\mathcal{P}$ consists of products of the marginal beliefs in the agents' ranges; it is not in general the convex hull of the agents' full joint priors. For outcomes with more than two possible values per coordinate, (RANGE) retains the stated marginal bounds without asserting equality to a marginal convex hull.

Zero-revenue outcomes are admitted by the weak aggregate definition; positive revenue gives the auctioneer a strict gain over no action without requiring strict surplus from any agent.

For the example in Section 3.1, no value is allocated and the transfers cancel at each decision in every outcome. Hence $\sum_i U_{i,t} = 0$ for every decision t, under every certifying prior, and the right-hand side of (CERT) is zero. The dynamic NBR criterion therefore excludes its positive revenue 2Λ, even though both agents satisfy own-belief individual rationality.

3.5 Why range constraints and the aggregate certificate are needed

Without the range restriction, a certificate could assign positive probability to success of a box that every agent regards as certain to fail, and use its value to justify arbitrarily large payments. Requiring that box's marginal to lie between the agents' corresponding marginal beliefs excludes this certificate.

Finally, prohibiting up-front transfers to agents does not bound benchmark revenue. Agents can reproduce the same bet through state-contingent transfers attached to different decisions while all up-front payments flow to the mechanism. The resulting revenue remains deterministic and unbounded under own-belief individual rationality. Appendix B gives this construction as well. The dynamic NBR definition requires the revenue certificate (CERT); restricting the signs of up-front transfers is insufficient.

4 Revenue Guarantees for Strategic Pandora

The two mechanisms face the same economic question. A high probability report can move a box forward and increase its chance of supplying the selected action, but the same report also determines an up-front payment. We ask whether competition forces those payments to recover a constant fraction of the optimal revenue among mechanisms in our admissibility class.

We formalize the comparison through two benchmarks, defined pointwise in the type profile. Let $\mathcal{A}_{\mathrm{K}}(p, v)$ and $\mathcal{A}_{\mathrm{RV}}(p, v)$ denote the admissible mechanism–equilibrium pairs under known and reported values, respectively. Thus $(\mathcal{M}, e)$ belongs to the relevant set if e is a pure equilibrium of $\mathcal{M}$ at (p, v) and its induced outcome satisfies dynamic NBR. Define

$$\mathrm{OPT}_{\mathrm{DNBR}}(p, v) := \sup_{(\mathcal{M},e)\in\mathcal{A}_{\mathrm{K}}(p,v)} \mathrm{Rev}_{\mathcal{M}}(e; p, v), \qquad \mathrm{OPT}_{\text{RV-DNBR}}(p, v) := \sup_{(\mathcal{M},e)\in\mathcal{A}_{\mathrm{RV}}(p,v)} \mathrm{Rev}_{\mathcal{M}}(e; p, v).$$

4.1 Known box values

Here $v_j > 0$ is public and agents report only success probabilities. We isolate the role of competition through one local condition. The profile has **constant probability competition** (PC-K) if every box has K distinct agents whose true success probabilities are at least an α fraction of the highest belief on that box. The required number of agents on each box depends on the discount factor and is independent of the number of boxes.

The mechanism creates the same bid-shading incentive as a first-price auction: without a credible rival, the most optimistic agent can lower its report and payment while retaining the reader role. A close probability rival creates a threshold: once revenue falls too low relative to the box's index, that rival can report just enough to take the role profitably. The sequential setting makes this argument substantially less local than in a single-item auction. Taking one reader role changes the inspection order, the probability of reaching every later box, and the payments attached to those boxes. The proof shows that these downstream effects can still be charged to the revenue already collected on the prefix.

Our exact dynamic NBR membership proof uses one additional selection property: every selected reader has a nonnegative position on its box. Appendix B states the equilibrium-selection rule that supplies this property, while Appendix C gives the revenue argument and exact constants.

Theorem 2 (Known-values comparison). *For every $0 < \alpha \leq 1$ and $0 < \delta < 1$, there is a finite probability depth $K_0(\delta)$, independent of the number of boxes, such that at every type profile satisfying* (PC-K_0)*, every existing pure RW equilibrium selected by the favorable membership tie rule satisfies*

$$\mathrm{Rev}_{\mathrm{RW}} = \Omega\Big(\alpha(1-\delta)^2 \,\mathrm{OPT}_{\mathrm{DNBR}}(p, v)\Big). \tag{KV-COMP}$$

The competition argument first gives an RW revenue lower bound for the broader pure-equilibrium class. The favorable membership selection is used separately to show that the induced outcome satisfies dynamic NBR.

The independence from the number of boxes is the substantive point. A longer system creates more possible reads, more order changes, and more opportunities for strategic reports to affect downstream reach. Nevertheless, the mechanism does not require a growing market on every box. Discounting limits the economically relevant prefix, and a constant local depth of probability competition is enough to control the entire inspection sequence.

Why competition is necessary. Without a close probability rival, RW revenue can be arbitrarily small relative to the dynamic NBR benchmark, even with two boxes. A simple voluntary direct mechanism has a pure-equilibrium outcome satisfying dynamic NBR whose revenue approaches the full value of the valuable box, whereas RW collects only the report needed to stay ahead of the next belief. As the second-highest belief falls relative to the highest, $\mathrm{Rev}_{\mathrm{RW}} / \mathrm{OPT}_{\mathrm{DNBR}}$ converges to zero. The familiar bid-shading obstruction of a first-price auction causes this failure through probability reports. Appendix E gives the construction and its exact tie-selection boundary.

Pure-equilibrium existence and the report domain. The maintained probability-report domain is $[0, 1]$, and the endpoint is substantive. On the open domain $[0, 1)$, a three-box instance has no pure ε-equilibrium below a fixed positive constant, under any complete tie rule, even though every true success probability is at most $3/5$ (and analogous constructions exist with arbitrarily small true probabilities). The failure is caused by a large ratio between agents' assessments of the probability of reaching later boxes. An agent that is more pessimistic about an early box assigns greater reach to later boxes than the adopted reports imply. It exploits this discrepancy by overreporting on the tail boxes. In the construction, reporting nearly one on B drives the payment for C to zero, while the agent's true beliefs still assign C positive reach, making the joint takeover profitable.

The closed domain removes that strict boundary deviation but does not settle universal existence. Coordinatewise report-raising repairs increase RW revenue and admit a strict repair potential, yet their limit need not be a Nash equilibrium: an order-changing deviation may lose on the promoted box while gaining on a pre-existing later role. This is why compactness and monotone repair are not enough. Appendices I and J give the exact results. Universal pure-equilibrium existence on $[0, 1]$ remains open.

Appendix D gives the separate ordinary mixed-equilibrium revenue bound. It is a revenue result and does not establish dynamic NBR membership for mixed equilibria.

4.2 Reported values

Dynamic NBR membership under reported values. RWSP separates the reader and writer roles. Establishing dynamic NBR membership therefore requires controlling the success transfer

paid by a writer distinct from the reader.

Theorem 3 (Writer-only no-overbidding). *At every pure equilibrium of RWSP in the interior probability-report domain, for every box j whose selected writer $w(j)$ differs from its reader $r(j)$, one has*

$$y_j \le v^j_{w(j)}. \tag{NO}$$

Appendix F proves Theorem 3.

When the reader and writer coincide, the success transfer is internal. Otherwise, Theorem 3 implies that the writer's net value upon success is nonnegative.

Theorem 4 (Reported-values dynamic NBR membership). *Every pure equilibrium of RWSP in the interior probability-report domain induces an outcome satisfying dynamic NBR.* Appendix F proves Theorem 4.

Why revenue needs two kinds of competition. Probability competition plays the same economic role on the reader side as under known box values: a low probability payment invites a sufficiently informed rival to take the role. It is no longer sufficient by itself. RWSP uses the second-highest value both as the success transfer and as the value in the box's index. If the highest-value agent has no close value rival, that second price can collapse even when the selected write is extremely valuable.

We therefore require **qualified value competition**: on every box, the second-highest true value is at least a β fraction of the highest true value. We also require pointwise exact **writer-allocation Envy-Free**: at the realized second price, the selected writer's true value is at least the price and every nonwriter's true value is at most the price. Qualified value competition keeps the price commensurate with the available value; writer-allocation Envy-Free connects the reported second-price allocation to the agents' true values. These are different requirements, and neither substitutes for probability competition on the reader side.

Theorem 5 (Reported-values comparison). *For every $0 < \alpha, \beta \le 1$ and $0 < \delta < 1$, there is a finite probability depth $K_0(\alpha, \beta, \delta)$, independent of the number of boxes, such that every existing pure equilibrium with interior probability reports that satisfies constant probability competition, qualified value competition, and pointwise exact writer-allocation Envy-Free satisfies*

$$\mathrm{Rev}_y = \Omega\Big(\alpha\beta(1-\delta)^3 \,\mathrm{OPT}_{\text{RV-DNBR}}\Big). \tag{SV-COMP}$$

By Theorem 4, every such equilibrium outcome satisfies dynamic NBR.

This result is not a direct extension of the known-values theorem. The probability report that moves a box forward, the value report that receives the write, and the second value report that sets the price may come from three different agents. The proof must therefore show simultaneously that probability competition supports the reader deviation, qualified value competition keeps the true second value commensurate with the highest value, and writer-allocation Envy-Free connects that true-value floor to the realized second price. Even with this separation, the required probability depth remains independent of the number of boxes. Appendix H gives the exact coefficient, the explicit K_0, and the sharper finite-horizon comparison.

4.3 Scope of the Revenue Guarantees.

The comparison guarantees in Theorems 2 and 5 apply only to equilibria in their stated classes. Universal pure-equilibrium existence for RW on $[0, 1]$ remains open, as does existence of a RWSP equilibrium satisfying exact writer-allocation Envy-Free. Appendix I discusses the RW existence boundary, and Appendix J proves nonexistence on the open report domain $[0, 1)$.

5 Related Work

Pandora, reservation indices, and learning. Weitzman (1979) derives an optimal inspection order and stopping rule for independent boxes with known prize distributions, inspection costs, and delay. The rule assigns a reservation value to each box. We use Bernoulli boxes with delay and zero inspection cost, and require search to stop at the first success. The ordering lemma gives the allocation rule used by RW. Agents report the probabilities used to order the boxes and determine payments.

Olszewski and Weber (2015) study when a Pandora rule is optimal for more general objectives and relate the rule to the Gittins index for an equivalent bandit problem. Doval (2018) allows selection without inspection, Boodaghians et al. (2020) impose order constraints, and Chawla et al. (2020, 2023) study correlated boxes. Esfandiari et al. (2019) study online Pandora problems and their relation to prophet and bandit problems. Atsidakou et al. (2022) learn contextual reservation values, including under bandit feedback. In our model, each box is inspected at most once per round; the mechanism takes probability reports as given and does not learn distributions from repeated observations. Several agents may report different beliefs about the same box.

Chawla et al. (2026) study posted inspection prices for one sequential searcher. Seller and buyer know the distributions, and the buyer pays for each inspection. In RW, agents report success probabilities and pay up front according to the reported objective term. Our benchmark maximizes deterministic revenue over mechanism–equilibrium pairs whose outcomes satisfy dynamic NBR.

Dynamic mechanism design and information acquisition. Parkes and Singh (2003) use an MDP to implement a long-run value-maximizing policy in Bayesian Nash equilibrium when agents arrive and depart. The dynamic pivot mechanism implements efficient allocation under a specified stochastic model (Bergemann and Välimäki 2010). Qiu et al. (2024) learn an unknown MDP over repeated episodes. These mechanisms evaluate policies under one known or learned model. Our agents may disagree about the same process, and we compare revenue without a common prior.

Cavallo and Parkes (2008) allocate costly deliberation in metadeliberation auctions. In the basic model, each agent reports an MDP for deliberation about its own value. Their extensions allow deliberation about other agents' values. Our mechanism orders inspections of shared conditions. Every player may have a private belief and a private value for every branch of the same process. Under reported values, the player reporting a condition probability need not be the player that values the resulting write.

Bergemann and Välimäki (2002) and Persico (2000) study information acquisition before a static allocation. Bergemann and Välimäki show that efficient mechanisms provide socially efficient incentives to acquire information in their private-value model. In Persico's model with affiliated information, a first-price auction gives stronger incentives for continuous signal acquisition than a second-price auction. Our model takes private beliefs and values as given. It does not model or price their acquisition.

Porter et al. (2008) elicit task-success probabilities and execution costs in fault-tolerant mechanism design. The probability concerns the agent assigned the task. In our model, several agents hold private beliefs about the same external condition.

Algorithms as mechanisms and equilibrium guarantees. Lucier and Syrgkanis (2015) and Dütting, Kesselheim, and Tardos (2021) obtain equilibrium welfare guarantees by combining greedy and relax-and-round allocation algorithms with payment rules. Hartline, Hoy, and Taggart (2014) obtain equilibrium revenue guarantees from the prices agents face when deviating. RW combines the discounted Weitzman order with the up-front objective-term payment rule. A probability report

changes a box's position and the probability of reaching later boxes. We bound revenue by certifiable allocated value through dynamic NBR and compare RW revenue with the resulting benchmark.

Heterogeneous beliefs and no-betting. Gilboa, Samuelson, and Schmeidler (2014) use a hypothetical shared belief to refine Pareto comparisons. Their strict-gain requirement applies to every affected agent. We include the auctioneer and allow weak gains for agents. Positive auctioneer revenue can supply the strict gain.

For a single decision, we restrict certification to the convex hull of agents' beliefs. Brunnermeier, Simsek, and Xiong (2014) use this set to classify belief-neutral efficiency and inefficiency, evaluating every belief in the set. Our certificate requires one belief in the set. NBR ranks admissible outcomes by deterministic auctioneer revenue; the dynamic NBR criterion extends the comparison to sequential decisions (Section 3). The aggregate certificates bound revenue by certifiable allocated value but may admit mechanisms containing bets. The dynamic NBR definition permits a different product prior at each decision, with marginals in the agents' belief ranges. The same prior evaluates payoffs at that decision and its probability of being reached.

Transaction sequencing and conditional execution. Exchanges and brokers, such as Nasdaq and on-chain order books such as Hyperliquid, accept conditional orders in a limited language and order them by price and then time (Harris 2003; Hyperliquid 2024). Uniswap v4 hooks provide a related form of conditional execution, activated by a separate user calling the pool. The calling user bears the gas cost of executing an earlier user's stored instruction (Adams et al. 2024). Frequent batch auctions instead process orders at discrete intervals (Budish, Cramton, and Shim 2015). These approaches do not extend naturally to a composable environment where one rule may include operations across several markets. No single price ranks a composite rule, rendering Price First ill-defined. Submission time is also meaningless when a rule may wait for days or weeks before its condition holds. We therefore focus on content-agnostic mechanisms, which use only the reported condition probability and write value, not the content and semantics of instructions.

Transaction fee mechanisms price access for an available set of transactions (Roughgarden 2021; Chung and Shi 2023). Verifiable Sequencing Rules restrict how an operator orders included exchange transactions and use transaction direction and exchange-state price effects (Ferreira and Parkes 2023). We allocate condition reads and the next transaction slot among standing instructions, using only reported probabilities and write values.

Chitra, Ferreira, and Kulkarni (2023) use blockchains to restrict deviations by a self-interested auctioneer. We assume execution of the stated mechanism after admission. Event-driven contracts, active databases, keepers, and solvers already evaluate conditions and dispatch actions (Widom and Ceri 1996; Paton and Díaz 1999; Kaleem et al. 2021; Zhao et al. 2022).

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Appendices

Appendix A: Discounted Pandora Ordering

Lemma 1 (Discounted Weitzman optimality). *Among policies that execute the first successful box and then stop, ordering the boxes in weakly decreasing order of z_j maximizes W_δ.*

Proof. Consider two adjacent boxes a, b, followed by a continuation of conditional value C. The contribution of the continuation is the same under the orders a, b and b, a, because in both cases it is multiplied by $\delta^2(1-p_a)(1-p_b)$. Subtracting the remaining terms gives

$$\begin{aligned} W_{ab} - W_{ba} &= p_a v_a(d + \delta p_b) - p_b v_b(d + \delta p_a) \\ &= (d + \delta p_a)(d + \delta p_b)\left(\frac{p_a v_a}{d + \delta p_a} - \frac{p_b v_b}{d + \delta p_b}\right), \end{aligned}$$

where $d := 1 - \delta$. Thus a weakly precedes b exactly when $z(p_a, v_a) \geq z(p_b, v_b)$. Repeated adjacent swaps prove that decreasing index order maximizes W_δ. Since the Bernoulli model awards the execution value of the first successful box and later inspection only delays that award, search terminates at the first success. This proves the lemma.

Appendix B: Dynamic No-Betting Revenue: Certificates, Caps, and Known-Values Membership

B.1 Tie rules and equilibrium selection

An **admissible report-blind tie rule** is deterministic. It may depend on true types, public values, agent and box identities, tied-agent sets, and selected agent–box combinations, but not on the numerical level shared inside a tie. The **favorable membership tie rule** is the narrower selection among admissible rules for which the boxwise deviation argument rules out a negative selected probability position at every selected pure equilibrium. This is the selection used in the known-values dynamic NBR membership statement. For an implementation, fixed priorities and tie rules must depend only on recorded or public inputs; the broader type-dependent class is used only for analytical equilibrium selection.

B.2 Exact Strategic Pandora certificates

The decision-by-decision individual inequalities extend the motivation from the no-action comparison including the auctioneer in Section 3.2. Both membership proofs establish this stronger route to the aggregate dynamic NBR requirement by choosing $\nu^t \in \mathcal{P}$ for each decision t so that $\sum_t \mathbb{E}_{\nu^t}[U_{i,t}] - t_i \geq 0$ for every agent i. Summing over agents implies (CERT). The benchmark needs only (CERT), which bounds revenue by certifiable aggregate allocation. For $R > 0$, the auctioneer already has a strict gain over no action, without requiring strict certified surplus from any agent; $R = 0$ is admitted by the weak aggregate definition.

Fix an outcome order π. Put

$$\underline{p}_k := \min_i p_i^k, \qquad \overline{p}_k := \max_i p_i^k.$$

Every box-specific product prior ν^j whose marginal probabilities θ_k^j lie in $[\underline{p}_k, \overline{p}_k]$ satisfies

$$L_j \leq J_j(\nu^j) \leq U_j, \tag{SA4}$$

where

$$L_j := \delta^{t(j)-1} \prod_{k \prec_\pi j} (1 - \overline{p}_k) \underline{p}_j, \qquad U_j := \delta^{t(j)-1} \prod_{k \prec_\pi j} (1 - \underline{p}_k) \overline{p}_j.$$

Indeed, each preceding failure factor lies between $1 - \overline{p}_k$ and $1 - \underline{p}_k$, and the target success factor lies between $\underline{p}_j$ and $\overline{p}_j$. Multiplying the nonnegative inequalities gives both exact edges.

B.3 Why the Dynamic NBR Restrictions Are Needed

The examples below use boundary beliefs to show why dynamic NBR restricts certifying beliefs to the marginal ranges, permits a different certifying prior at each decision, and imposes the aggregate revenue certificate. The examples technically lie outside the interior true-belief domain that is used in the main theorems, but the boundary beliefs make the role of each restriction clear.

B.3.1 Unrestricted certifying beliefs

Without the range restriction, a certificate can rely on an event that every agent regards as impossible. Consider two agents, two boxes ordered A, B, a discount factor $0 < \delta < 1$, and $\Lambda > 1$. Let

$$v_A = \Lambda + 1, \qquad v_B = 1,$$

and let the agents' beliefs be

$$(p_1^A, p_1^B) = (0, 1), \qquad (p_2^A, p_2^B) = (0, 0).$$

Allocate the value of each successful box to agent 1, charge $t_1 = \Lambda$ and $t_2 = 0$, and set

$$\beta_{1B} = -\frac{\Lambda}{\delta}, \qquad \beta_{2B} = \frac{\Lambda}{\delta}, \qquad \beta_{1A} = \beta_{2A} = 0.$$

If B is reached and succeeds, agent 2 pays Λ/δ to agent 1. Agent 1 expects the transfer, whereas agent 2 believes it will never occur. Their own-belief utilities are

$$U_1^{\text{own}} = \delta - \Lambda + \Lambda = \delta > 0, \qquad U_2^{\text{own}} = 0,$$

while deterministic revenue is $R = \Lambda$.

If the range condition is removed, a certifying prior can assign probability one to success of A, although both agents assign that event probability zero. The value $\Lambda + 1$ then certifies the up-front charge, and success of A prevents the contingent transfer on B from being reached. Certification and own-belief individual rationality hold for every Λ, so revenue is unbounded. The range restriction excludes exactly this fictitious source of value.

B.3.2 One prior for the complete outcome

A single range-constrained product prior for the entire sequential outcome is too restrictive. Let $\delta = 0.9$, $v_A = v_B = 1$, and consider four agents with beliefs

$$p_1 = p_2 = (0.5, 0), \qquad p_3 = p_4 = (0, 0.1).$$

The first pair reports $(0.5, 0)$, and the second reports $(0, 0.2)$. RW inspects A before B. The selected reader on A receives expected value 0.5 and pays 0.5. The reader on B believes that A certainly fails, so its expected discounted value and its up-front payment are both

$$0.9(1 - 0)(0.1) = 0.9(1 - 0.5)(0.2) = 0.09.$$

The report profile is a pure equilibrium and RW revenue is 0.59.

Any single product prior inside the agents' belief ranges has $0 \le \theta_A \le 0.5$ and $0 \le \theta_B \le 0.1$. Its expected allocated value is at most

$$\theta_A + 0.9(1-\theta_A)\theta_B \le 0.5 + 0.9(1-0.5)(0.1) = 0.545 < 0.59.$$

No single range-constrained prior can therefore certify the outcome. Decision-specific priors can: use $(\theta_A^A, \theta_B^A) = (0.5, 0)$ for decision A and $(\theta_A^B, \theta_B^B) = (0, 0.1)$ for decision B. Their certified masses are 0.5 and 0.09, exactly the mechanism revenue. The obstruction is reach. The reader on B is pessimistic about A and therefore assigns more probability to reaching B than a single coordinatewise optimistic prior does.

B.3.3 Prohibiting up-front transfers

Requiring all up-front payments to flow to the mechanism also fails to bound benchmark revenue. Let both boxes have value one, order them as A, B, and take two agents with beliefs

$$(p_1^A, p_1^B) = (1, 0), \qquad (p_2^A, p_2^B) = (0, 1).$$

Allocate no box value and charge $t_1 = t_2 = \Lambda$. Set

$$(\beta_{1A}, \beta_{2A}) = (-\Lambda, \Lambda), \qquad (\beta_{1B}, \beta_{2B}) = \left(\frac{\Lambda}{\delta}, -\frac{\Lambda}{\delta}\right).$$

The contingent payments cancel at each box, so retained revenue is the deterministic amount 2Λ. Under agent 1's belief, A succeeds and its contingent receipt offsets its up-front payment. Under agent 2's belief, A fails and the discounted receipt on B offsets its up-front payment. Both agents obtain zero own-belief utility, while revenue grows without bound. Because no value is allocated, the aggregate bound is zero and the outcome cannot enter the dynamic NBR benchmark. Thus a sign restriction on up-front payments is not sufficient to bound benchmark revenue. The aggregate certificate supplies that restriction without requiring an absence of betting in admitted mechanisms.

B.4 Revenue caps

The cap uses only aggregate certification. Contingent transfers between agents cancel decision by decision in every state, leaving aggregate allocation.

Put $p_j^{[1]} := \max_i p_i^j$ and $TW := W_\delta(p^{[1]}, v)$. For reported values, put $p_j^{\max} := \max_i p_i^j$, $v_j^{\max} := \max_i v_i^j$, and $W^{\max} := W_\delta(p^{\max}, v^{\max})$. Finally, put

$$G_M(\delta) := \sum_{t=1}^{M} \delta^{t-1} \le \frac{1}{1-\delta}.$$

Suppose first that values are public. The aggregate certificate and statewise feasibility imply the following, where V_{ij} is agent i's discounted allocation contribution from box j:

$$R \le \sum_j \sup_{\nu \in \mathcal{P}} \mathbb{E}_\nu \left[\sum_i V_{ij} \right] \le \sum_j U_j v_j.$$

If box $j = \pi_t$, its upper edge is at most $\delta^{t-1} p_j^{[1]}$. Hence

$$R \le \sum_{t=1}^{M} \delta^{t-1} p_{\pi_t}^{[1]} v_{\pi_t} \le \sum_{t=1}^{M} \delta^{t-1} TW = G_M(\delta) TW.$$

Here the second inequality follows because placing any box j first in the true-Weitzman objective gives $p_j^{[1]} v_j \le TW$. Taking the appropriate supremum over pure equilibria whose outcomes satisfy dynamic NBR and over mechanisms proves

$$\mathrm{OPT}_{\mathrm{DNBR}} \le G_M(\delta) TW. \tag{B.1}$$

Under reported values, statewise aggregate allocation on success of box j is at most $v_j^{\max}$. Repeating the proof gives

$$\mathrm{OPT}_{\text{RV-DNBR}} \le G_M(\delta) W^{\max}. \tag{B.2}$$

B.5 Exact membership under known box values

Fix an existing pure RW equilibrium selected by the favorable membership tie rule. Let

$$m_j := D_j(q)\widehat{q}_j$$

be the mass priced on box j. The favorable rule gives the selected reader a nonnegative box position:

$$D_{r(j),j} p^j_{r(j)} - m_j \ge 0. \tag{B.3}$$

The reader's true mass is at most U_j, so (B.3) gives $m_j \le U_j$.

For each box j, choose the product prior

$$\theta^j_k = \begin{cases} \overline{p}_j, & k = j, \\ \underline{p}_k, & k \ne j. \end{cases} \tag{B.4}$$

Every marginal lies in its coordinatewise range, and $J_j(\nu^j) = U_j$. Since $L_j \le U_j$, this certificate satisfies both exact edges of (SA4), while $J_j(\nu^j) \ge m_j$.

Agent i's up-front payment is

$$t_i = \sum_{j : r(j) = i} m_j v_j.$$

Under the box-specific certificates,

$$\sum_j \mathbb{E}_{\nu^j}[V_{ij}] - t_i = \sum_{j : r(j) = i} (U_j - m_j) v_j \ge 0. \tag{B.5}$$

Thus the stronger agent-by-agent certificate holds. Equilibrium opt-out gives own-belief individual rationality. RW has no terminal inter-agent transfers, its revenue is deterministic, and its aggregate allocation equals the produced value statewise. Equations (B.3)–(B.5) prove the known-values membership claim in Section 4.1.

Appendix C: Pure RW Revenue under Constant Probability Competition

Theorem 2 (Known-values comparison). *For every $0 < \alpha \le 1$ and $0 < \delta < 1$, there is a finite probability depth $K_0(\delta)$, independent of the number of boxes, such that at every type profile satisfying* (PC-K_0)*, every existing pure RW equilibrium selected by the favorable membership tie rule satisfies*

$$\mathrm{Rev}_{\mathrm{RW}} = \Omega\Big(\alpha(1-\delta)^2 \,\mathrm{OPT}_{\mathrm{DNBR}}(p, v)\Big). \tag{KV-COMP}$$

Proof. Fix an existing pure RW equilibrium with probability reports in $[0,1]$ and an admissible report-blind tie rule as defined in Appendix B.1. Under (PC-K_0), the intermediate revenue inequalities below apply to this equilibrium. For equilibria selected by the favorable membership tie rule, the final inequality gives the comparison with dynamic NBR.

Put $d := 1-\delta$, $p_j^{[1]} := \max_i p_i^j$, $TW := W_\delta(p^{[1]}, v)$, and define

$$Z_i := \max_j z(p_i^j, v_j), \qquad Z^* := \max_j z(p_j^{[1]}, v_j).$$

Let $Z^{[r]}$ be the r-th largest Z_i, with unavailable ranks set to zero. For $1 \le L \le M$, put

$$\tau_L := \begin{cases} \delta^L, & L < M, \\ 0, & L = M. \end{cases}$$

C.1 Two bounds

For every fixed order π,

$$W_\delta(p_i, v; \pi) \le Z_i, \qquad TW \le Z^*, \qquad \max_i Z_i = Z^*. \tag{C.1}$$

For the first inequality, work backward through π. If a suffix has value at most Z_i, adding a box of probability p and value v gives at most

$$pv + \delta(1-p)Z_i \le (d+\delta p)Z_i + \delta(1-p)Z_i = Z_i.$$

The second inequality is identical with the boxwise maximum probabilities, and the third follows by interchanging the two maxima.

C.2 The position inequality

Lemma C.1. For every $0 \le L \le M$, with $\tau_0 := 1$,

$$\mathrm{Rev}_{\mathrm{RW}} \ge (d - \tau_L)_+ Z^{[L+1]}. \tag{C.2}$$

Proof. At most L distinct agents own reader roles in the first L equilibrium positions. Among the $L+1$ agents with the largest Z_i's, choose h who owns none of these roles, so $Z_h \ge Z^{[L+1]}$. Let box B attain Z_h, and let $\widehat{Z}$ be the leading reported index. The first payment gives

$$\mathrm{Rev}_{\mathrm{RW}} \ge d\widehat{Z}. \tag{C.3}$$

When $\widehat{Z} \ge Z_h$, this proves (C.2). Otherwise, all reader roles of h lie after position L. Removing other agents' roles and closing the resulting gaps, then applying (C.1), gives

$$U_h \le \tau_L Z_h. \tag{C.4}$$

Agent h sets every non-target report to zero and reports on B just above the level q satisfying $z(q, v_B) = \widehat{Z}$. Since $z(p_h^B, v_B) = Z_h > \widehat{Z}$, we have $q < p_h^B$, and the target gain approaches

$$(p_h^B - q)v_B \ge d(Z_h - \widehat{Z}). \tag{C.5}$$

Every role retained at report zero is nonnegative. The Nash inequality, (C.4), and (C.5) give

$$d(Z_h - \widehat{Z}) \le \tau_L Z_h.$$

Adding (C.3) proves (C.2). For $L = M$, agent h owns no role and the same argument uses $\tau_M = 0$. □

C.3 Using constant probability competition

Let B^* attain Z^*, and choose the K agents supplied by (PC-K) on B^*. Since

$$z(\alpha p, v) \geq \alpha z(p, v),$$

each of these agents has $Z_i \geq \alpha Z^*$. Thus, for $1 \leq L \leq \min\{M, K-1\}$,

$$Z^{[L+1]} \geq \alpha Z^*. \tag{C.6}$$

Equations (C.1), (C.2), and (C.6) imply

$$\mathrm{Rev}_{\mathrm{RW}} \geq Q^{\mathrm{PC}}_{M,K}(\alpha, \delta) TW,$$

where

$$Q^{\mathrm{PC}}_{M,K}(\alpha, \delta) := \alpha \max_{1 \leq L \leq \min\{M, K-1\}} (d - \tau_L)_+. \tag{C.7}$$

For the finite-M companion bound, define

$$C_1 := \frac{1}{\alpha}, \qquad C_t := \max\left\{ C_{t-1} + \frac{2}{\alpha d}, \left(1 + \frac{2}{\alpha}\right) C_{t-1} \right\}. \tag{C.8}$$

If the equilibrium order is $(j_1, \ldots, j_M)$, put

$$E_t := p^{[1]}_{j_t} v_{j_t}, \qquad A_t := \sum_{s=1}^{t} E_s.$$

A current nonreader on j_t has probability at least $\alpha p^{[1]}_{j_t}$. If the report needed to move j_t first is infeasible or at least half this probability, the first payment bounds E_t by $2\,\mathrm{Rev}_{\mathrm{RW}}/(\alpha d)$. Otherwise the deviation creates target utility at least $\alpha E_t/2$, so the preceding prefix must have value at least that amount. Induction using the two cases in (C.8) gives

$$A_t \leq C_t \,\mathrm{Rev}_{\mathrm{RW}}\,. \tag{C.9}$$

Consequently

$$\mathrm{Rev}_{\mathrm{RW}} \geq \tilde{c}^{\mathrm{PC}}_{M,K}(\alpha, \delta) TW, \qquad \tilde{c}^{\mathrm{PC}}_{M,K} := \max\left\{ \frac{1}{C_M}, Q^{\mathrm{PC}}_{M,K} \right\}. \tag{C.10}$$

C.4 Constant depth and the restricted-class comparison

Define

$$L_0 := \min\{\ell \geq 1 : \delta^\ell \leq d/2\}, \qquad K_0 := \max\{3, L_0 + 1\}. \tag{C.11}$$

Choose $L = \min\{M, L_0\}$ in (C.7). Then $L + 1 \leq K_0$, and either $L = M$, so $\tau_L = 0$, or $\tau_L \leq d/2$. Therefore every pure equilibrium under (PC-K_0) satisfies

$$\mathrm{Rev}_{\mathrm{RW}} \geq \frac{\alpha(1-\delta)}{2} TW. \tag{C.12}$$

Combining (C.12) with (B.1) proves

$$\mathrm{Rev}_{\mathrm{RW}} \geq \frac{\alpha(1-\delta)}{2G_M(\delta)}\,\mathrm{OPT}_{\mathrm{DNBR}} \geq \frac{\alpha(1-\delta)^2}{2}\,\mathrm{OPT}_{\mathrm{DNBR}}\,. \tag{C.13}$$

Equation (C.12) is the exact intermediate revenue bound. For equilibria selected by the favorable membership tie rule, (C.13) is the exact form of Theorem 2.

Appendix D: Ordinary Mixed RW Revenue

We show that ordinary mixed RW equilibria under known box values satisfy a configuration-dependent lower bound on expected revenue. Under constant probability competition with a number of agents depending only on δ, expected revenue is an $\Omega(\alpha(1-\delta)^2)$ fraction of the nonstrategic benchmark.

Fix an ordinary mixed RW equilibrium $\sigma = \prod_i \sigma_i$ under known box values. Put $d := 1-\delta$, $p_j^{[1]} := \max_i p_i^j$, $TW := W_\delta(p^{[1]}, v)$, $Z_i := \max_j z(p_i^j, v_j)$, and $Z^* := \max_j z(p_j^{[1]}, v_j)$. Let $Z^{[r]}$ be the r-th largest Z_i, and put $\tau_L := \delta^L$ for $L < M$ and $\tau_M := 0$. Let

$$R := \mathbb{E}_\sigma[\mathrm{Rev}_{\mathrm{RW}}],$$

let u_i be agent i's equilibrium expected utility, and put $c_0 := 1 - e^{-1}$. The all-zero report gives $u_i \geq 0$.

Fix an agent i and target box B. Let

$$Z := z(p_i^B, v_B), \qquad T_i(q_{-i}) := \max_j z\left(\max_{h \neq i} q_h^j, v_j\right).$$

For every realized profile, the first payment gives $\mathrm{Rev}_{\mathrm{RW}} \geq dT_i$, hence

$$R \geq d\,\mathbb{E}[T_i]. \tag{D.1}$$

For a fixed $0 < x < Z$, let $q(x)$ satisfy $z(q(x), v_B) = x$. The fixed deviation reports $q(x)$ on B and zero elsewhere. Since $x < Z$, strict monotonicity of $z(\cdot, v_B)$ gives $q(x) < p_i^B$. On $\{T_i < x\}$, box B is first and agent i is its reader. The deviation utility on this event is at least

$$(p_i^B - q(x))v_B \geq d(Z - x).$$

On $\{T_i \geq x\}$, the deviation utility is nonnegative if B is first. If B is later, its payment is at most $\delta q(x) v_B \leq \delta x \leq \delta T_i$. Hence the deviation utility is bounded below by

$$d(Z-x)\mathbf{1}_{\{T_i < x\}} - \delta T_i.$$

The Nash inequality and (D.1) give

$$u_i + \frac{\delta}{d} R \geq d(Z-x)\Pr(T_i < x). \tag{D.2}$$

Put $C := R/d$ and

$$a := \frac{u_i + \delta C}{dZ}.$$

If $a \geq 1$, then $(u_i + C)/(dZ) \geq 1$. Suppose $0 < a < 1$. Equation (D.2) gives

$$\Pr(T_i < x) \leq \frac{aZ}{Z-x} \qquad (0 < x < Z).$$

Using (D.1) and integrating the corresponding lower bound on $\Pr(T_i \geq x)$ over $0 \leq x \leq (1-a)Z$ gives

$$\frac{C}{Z} \geq 1 - a - a\log(1/a).$$

If $a = 0$, equation (D.2) gives $\Pr(T_i < x) = 0$ for every $0 < x < Z$, so $C/Z \geq 1$ by (D.1). For $0 < a < 1$,

$$\frac{u_i + C}{dZ} = a + \frac{C}{Z}$$

and $a \log(1/a) \leq e^{-1}$. The cases $a = 0$ and $a \geq 1$ give the same lower bound directly. Therefore

$$u_i + \frac{R}{d} \geq c_0 d\, z(p_i^B, v_B). \tag{D.3}$$

Choose for each agent a box attaining $Z_i = \max_j z(p_i^j, v_j)$. Applying (D.3) to this box gives

$$u_i + \frac{R}{d} \geq c_0 d Z_i.$$

Fix $1 \leq K \leq n$ and $1 \leq L \leq M$, and let H_K contain the K agents with the largest Z_i's. For every realized report profile with induced order π, agent i's gross payoff is at most $W_\delta(p_i, v; \pi) \leq Z_i$, by the first bound in Appendix C. If agent i owns no reader role in the first L positions, its gross payoff is at most $\tau_L Z_i$. At most L agents in H_K own a reader role in these positions. Up-front payments are nonnegative. Summing the resulting pointwise bounds and taking expectations gives

$$\sum_{i \in H_K} \frac{u_i}{Z_i} \leq K\tau_L + (1 - \tau_L)L. \tag{D.4}$$

Divide (D.3) by Z_i, sum over H_K, and use $Z_i \geq Z^{[K]}$. This proves

$$R \geq d \left(c_0 d - \tau_L - (1 - \tau_L)\frac{L}{K} \right)_+ Z^{[K]}. \tag{D.5}$$

Equation (D.5) is the configuration-dependent ordinary mixed bound.

Assume the constant probability competition condition (PC-K) from Section 4.1. Let B^* attain Z^*, and choose the K agents supplied by the condition on B^*. Since

$$z(\alpha p, v) \geq \alpha z(p, v),$$

each of these agents has $Z_i \geq \alpha Z^*$. Hence $Z^{[K]} \geq \alpha Z^* \geq \alpha TW$, and (D.5) gives, for every $1 \leq L \leq M$,

$$R \geq \alpha d \left(c_0 d - \tau_L - (1 - \tau_L)\frac{L}{K} \right)_+ TW. \tag{D.6}$$

Define

$$L_0^{\mathrm{mix}} := \min \left\{ \ell \geq 1 : \delta^\ell \leq \frac{c_0 d}{4} \right\}, \qquad K_0^{\mathrm{mix}} := \left\lceil \frac{4L_0^{\mathrm{mix}}}{c_0 d} \right\rceil. \tag{D.7}$$

Under (PC-K_0^{mix}), choose $L = \min\{M, L_0^{\mathrm{mix}}\}$. Then $\tau_L \leq c_0 d/4$ and $L/K_0^{\mathrm{mix}} \leq c_0 d/4$, so (D.6) gives

$$R \geq \frac{c_0}{2} \alpha d^2 TW = \Omega\Big(\alpha (1 - \delta)^2 TW\Big). \tag{D.8}$$

Equation (D.8) establishes the expected-revenue bound for ordinary mixed equilibria. The paper's dynamic NBR membership results apply to pure equilibria.

Appendix E: A Two-Box Separation

We show that RW revenue can become arbitrarily small relative to the dynamic NBR benchmark in a two-box instance when one box lacks a close probability rival.

Fix $0 < \delta < 1$, put $d := 1 - \delta$, and let $r > 0$ tend to zero. There are two boxes A, B and two agents with

$$p_1 = (1, 1), \qquad p_2 = (r, 1), \qquad v_B = 1.$$

Set

$$\kappa := d + \delta r, \qquad v_A := \frac{\kappa^2}{r(d + \delta\kappa)}. \tag{E.1}$$

At the report profile

$$q_1 = (r, 0), \qquad q_2 = (r, 0), \tag{E.2}$$

select agent 1 on A, agent 2 on B, and inspect A first.

Agent 1's utility is $(1 - r)v_A$. Lowering its report on A loses this role, raising it increases its payment, and a positive report on B that leaves B second has zero subjective reach. Moving B first requires a report at least κ, because

$$\frac{rv_A}{d + \delta r} = \frac{\kappa}{d + \delta\kappa}.$$

Such a deviation yields utility at most one, whereas $(1 - r)v_A > 1$ for all sufficiently small r.

Agent 2's utility is $\delta(1 - r)$. If B remains second and agent 2 does not win A, a positive report b on B gives at most $\delta(1 - r)(1 - b)$. If it also wins A, its report there must satisfy $a > r$, and its total utility is at most

$$\delta(1 - r) + (r - a)v_A - \delta(1 - a)b \le \delta(1 - r).$$

Moving B first requires a report at least κ, and gives utility at most

$$1 - \kappa = \delta(1 - r).$$

If agent 2 also wins the later A, its subjective reach there is zero because it believes B succeeds surely, while its payment is nonnegative. The analogous later-box reach is zero for agent 1. Thus these cases also cover simultaneous two-coordinate deviations, and (E.2) is a pure equilibrium for every sufficiently small r.

RW revenue at this equilibrium is rv_A. For any $\lambda < 1$, consider the voluntary direct mechanism that inspects only A, allocates its value to agent 1, and charges λv_A if agent 1 opts in. The agent obtains positive utility. For its box-A certificate, take the product prior whose A-marginal is 1; choose every unused marginal within its coordinatewise range. Its mass lies between the exact edges r and 1, and the agent-by-agent certificate utility is $(1 - \lambda)v_A \ge 0$. Revenue is deterministic and allocation is feasible. The outcome therefore has exact dynamic NBR membership. Taking the supremum as $\lambda \uparrow 1$ gives

$$\mathrm{OPT}_{\mathrm{DNBR}} \ge v_A.$$

Consequently

$$\frac{\mathrm{Rev}_{\mathrm{RW}}}{\mathrm{OPT}_{\mathrm{DNBR}}} \le r = \frac{p_A^{[2]}}{p_A^{[1]}} \longrightarrow 0.$$

This proves the two-box separation discussed in Section 4.1.

Appendix F: Writer-Only No-Overbidding and Reported-Values Membership

Theorem 3 (Writer-only no-overbidding). *At every pure equilibrium of RWSP in the interior probability-report domain, for every box j whose selected writer $w(j)$ differs from its reader $r(j)$, one has*

$$y_j \le v^j_{w(j)}. \tag{NO}$$

Proof. Fix a pure equilibrium in the interior probability-report domain of Theorem 3, a box A, and a selected writer $w \neq r(A)$. Write

$$y := y_A, \qquad \widehat{q} := \widehat{q}_A, \qquad H := \{i \neq w : b^A_i \ge y\}.$$

Suppose for contradiction that $y > v^A_w$. The writer contribution of A to w's utility is strictly negative:

$$S_{w,A}(p_w;\pi)p^A_w(v^A_w - y) < 0. \tag{F.1}$$

We show that w can surrender this role while preserving the second price, adopted probability, complete index vector, and order.

If $|H| \ge 2$, lower b^A_w below y. Two other reports preserve the second price, all probability reports remain fixed, and the reader remains fixed. The negative term (F.1) disappears.

It remains to consider $H = \{s\}$. Then

$$b^A_s = y, \qquad b^A_k < y \quad (k \notin \{w, s\}). \tag{F.2}$$

If $q^A_w < q^A_s$, setting $b^A_w = y$ makes s the writer while preserving all other relevant quantities. If $q^A_w = q^A_s < \widehat{q}$, additionally lower q^A_w by an arbitrarily small amount; another agent preserves the adopted probability, while the effect on later personalized payments vanishes with the perturbation.

Suppose $q^A_s < q^A_w < \widehat{q}$. While q^A_w stays below $\widehat{q}$, define the nonnegative coefficient

$$K^y_w(A;\pi) := \sum_{\substack{j: A \prec_\pi j \\ r(j)=w}} \delta^{t_\pi(j)-1} \prod_{\substack{k \prec_\pi j \\ k \neq A}} (1 - q^k_w) q^j_w y_j. \tag{F.3}$$

If (F.3) is positive, a small increase in q^A_w is already profitable. If it is zero, set $b^A_w = y$ and $q^A_w = 0$. Agent s becomes the writer, another agent preserves $\widehat{q}$, and (F.3) shows that the finite probability change has no later-payment cost.

Finally, if $q^A_w = \widehat{q}$, the reader rule and $w \neq r(A)$ imply

$$r(A) = s, \qquad b^A_w = b^A_s = y, \qquad q^A_w = q^A_s = \widehat{q}.$$

Lower q^A_w slightly. Agent s remains reader and becomes writer; all indices and the order remain fixed, and the later-payment effect vanishes with the perturbation. Every case removes the fixed negative term (F.1) at arbitrarily small or zero other cost, contradicting equilibrium. This proves Theorem 3.

Theorem 4 (Reported-values dynamic NBR membership). *Every pure equilibrium of RWSP in the interior probability-report domain induces an outcome satisfying dynamic NBR.*

Proof. Fix the same pure equilibrium, and put $v^{\max}_j := \max_i v^j_i$. Let

$$m_{i,j} := \delta^{t(j)-1} \prod_{k \prec j} (1 - p^k_i) p^j_i$$

be agent i's true discounted reach-and-success mass. With the notation of Appendix B, coordinate-wise comparison gives

$$L_j \le m_{i,j} \le U_j \qquad \text{for every } i, j. \tag{F.4}$$

Choose for each j the upper-endpoint product prior (B.4), so $J_j(\nu^j) = U_j$.

Combine agent i's success terms on j into

$$e_{i,j} := \begin{cases} y_j, & i \text{ is only the reader,} \\ v_i^j - y_j, & i \text{ is only the writer,} \\ v_i^j, & i \text{ holds both roles,} \\ 0, & i \text{ holds neither role.} \end{cases} \tag{F.5}$$

The no-overbidding argument above handles a writer distinct from the reader, and the internal-transfer identity handles an agent with both roles. Hence

$$0 \le e_{i,j} \le v_j^{\max}. \tag{F.6}$$

Let t_i be agent i's total up-front reader payment. Its own-belief utility is

$$u_i = \sum_j m_{i,j} e_{i,j} - t_i.$$

The all-zero report is an opt-out deviation and gives nonnegative utility, so $u_i \ge 0$. Under the certificates,

$$u_i^{\text{cert}} = \sum_j U_j e_{i,j} - t_i \ge \sum_j m_{i,j} e_{i,j} - t_i = u_i \ge 0. \tag{F.7}$$

Thus the agent-by-agent certificate holds. Writer-to-reader transfers cancel statewise, retained revenue is the sum of the up-front payments, and aggregate allocation on box j is at most $v_j^{\max}$. Equations (F.4)–(F.7) prove Theorem 4. Notice that exact writer-allocation Envy-Free is nowhere used.

Appendix G: Ordinary Mixed Revenue for the Second-Price Mechanism

We derive expected-revenue bounds for ordinary mixed equilibria of RWSP that satisfy exact writer-allocation Envy-Free at almost every realized report profile. The bounds include a common guarantee under constant probability competition.

Put

$$d := 1 - \delta, \qquad c_\delta := \min_{\substack{0<a\le 1 \\ a \ge \delta[1 - a\log(1/a)]}} [1 - a\log(1/a)]. \tag{G.1}$$

The inequality $a\log(1/a) \le e^{-1}$ gives $c_\delta \ge 1 - e^{-1}$.

Assume finitely many agents and boxes, $0 < p_i^j < 1$, and finite $v_i^j \ge 0$. Probability and value reports range over $[0, 1]$ and $[0, \infty)$, respectively, so every full-report deviation below is admissible. Put $p_j^{\max} := \max_i p_i^j$, $v_j^{\max} := \max_i v_i^j$, let $v_j^{(2)}$ be the second-highest true value on box j, and put $W^{(2)} := W_\delta(p^{\max}, v^{(2)})$. Fix an ordinary mixed equilibrium of the mechanism and deterministic role, second-price, and tie rules in Section 2.3. Assume exact writer-allocation Envy-Free at almost every

realized report profile. Let R be expected up-front reader-payment revenue and u_i the expected utility of agent i. Fix a target box B and a set H of $K \geq 3$ agents. Write

$$x_i(B) := z(p_i^B, v_B^{(2)}), \qquad r_i(B) := p_i^B v_B^{(2)}. \tag{G.2}$$

Let $g_i(B)$ be the probability that replacing only agent i's value report on B by zero lowers the second price below $v_B^{(2)}$. Pointwise Envy-Free gives $y_B \geq v_B^{(2)}$, so at least two realized value reports reach this floor. Therefore

$$\sum_{i \in H} g_i(B) \leq 2. \tag{G.3}$$

Order the $x_i(B)$'s within H, and let $A_x(B,H)$ be their sum after deleting the two largest. Define $A_r(B,H)$ analogously. For $A, U \geq 0$, put

$$\Psi_\delta(A,U) := \sup_{s>0} \frac{[(1-e^{-s})A - sU]_+}{1+s\delta}. \tag{G.4}$$

G.1 The equilibrium-specific bound

Theorem G.1. For every B and H,

$$R \geq \max \left\{ \frac{d}{K} \left[c_\delta d \sum_{i \in H} (1 - g_i(B)) x_i(B) - \sum_{i \in H} u_i \right]_+ , \right.$$
$$\left. d \sup_{(s_i) \in (0,\infty)^H} \frac{[\sum_{i \in H} (1 - e^{-s_i})(1 - g_i(B)) r_i(B) - \sum_{i \in H} s_i u_i]_+}{K + \delta \sum_{i \in H} s_i} \right\}. \tag{M1}$$

In particular,

$$R \geq \frac{d}{K} \max \left\{ \left[c_\delta d A_x(B,H) - \sum_{i \in H} u_i \right]_+ , \Psi_\delta \left(A_r(B,H), \sum_{i \in H} u_i \right) \right\}. \tag{M1c}$$

Proof. Fix $i \in H$. Replace both reports of i by zero on every box, and let T_i be the largest remaining index. Restoring i's reports weakly raises every index. The first full-profile payment is therefore at least dT_i, so

$$\mathbb{E}[T_i] \leq \frac{R}{d} =: C. \tag{G.5}$$

Let G_i be the event that removal of i's target value report preserves the floor $v_B^{(2)}$, and put $g_i = \Pr(G_i^c)$.

For $0 < x < x_i(B)$, let $q_i(x)$ satisfy $z(q_i(x), v_B^{(2)}) = x$. Consider the fixed deviation that reports zero everywhere except for probability $q_i(x)$ and value zero on B. On $G_i \cap \{T_i < x\}$, box B is uniquely first, i is its unique reader, and its target utility is at least

$$(p_i^B - q_i(x)) v_B^{(2)} \geq d(x_i(B) - x).$$

If the target is later, its payment is at most δT_i. The Nash inequality therefore gives

$$u_i + \delta C \geq d(x_i(B) - x) \Pr(G_i, T_i < x). \tag{G.6}$$

Set $h_i := 1 - g_i$. From $\Pr(G_i, T_i < x) \geq \Pr(T_i < x) - g_i$, integrate the implied upper bound on $\Pr(T_i < x)$ up to the point where it becomes vacuous. The layer-cake identity and (G.5) yield

$$u_i + C \geq c_\delta d(1 - g_i)x_i(B). \tag{G.7}$$

For completeness, after normalizing

$$a := \frac{u_i + \delta C}{d(1 - g_i)x_i(B)}, \qquad r := \frac{u_i + C}{d(1 - g_i)x_i(B)},$$

the integration gives $r \geq 1 - a\log(1/a)$, while $u_i \geq 0$ gives $a \geq \delta r$. Definition (G.1) then implies $r \geq c_\delta$.

The second deviation draws the target probability report Q_i from $[0, (1 - e^{-s})p_i^B]$ with density

$$f_{i,s}(q) = \frac{1}{s(p_i^B - q)}, \tag{G.8}$$

reports value zero on B, and reports zero elsewhere. Conditional on G_i, integrating the first-position reader margin above the reduced-index threshold gives

$$su_i + (1 + s\delta)C \geq (1 - e^{-s})(1 - g_i)r_i(B). \tag{G.9}$$

A target that cannot become first makes the right side of the underlying pointwise bound nonpositive, so (G.9) also covers the boundary cases.

Sum (G.7) over H, sum (G.9) using arbitrary $s_i > 0$, and substitute $C = R/d$. This proves (M1). Equation (G.3) implies

$$\sum_{i \in H}(1 - g_i)x_i(B) \geq A_x(B, H), \qquad \sum_{i \in H}(1 - g_i)r_i(B) \geq A_r(B, H).$$

Using a common $s_i = s$ and optimizing proves (M1c). □

G.2 The type-dependent bound

Define

$$H_i := W_\delta(p_i, v^{\max}),$$

and, for $1 \leq L \leq M$,

$$T_{i,L}^{\text{tail}} := \max_{\pi} \sum_{t=L+1}^{M} \delta^{t-1} \left(\prod_{r<t}(1 - p_i^{\pi_r}) \right) p_i^{\pi_t} v_{\pi_t}^{\max}, \qquad \Delta_{i,L} := H_i - T_{i,L}^{\text{tail}}. \tag{G.10}$$

At most $2L$ distinct reader-or-writer identities occur in the first L positions. If $\Delta_{H,L}^{[\ell]}$ denotes the ℓ-th largest $\Delta_{i,L}$ among $i \in H$, put

$$\mathcal{U}_H(L) := \sum_{i \in H} T_{i,L}^{\text{tail}} + \sum_{\ell=1}^{\min\{2L,K\}} \Delta_{H,L}^{[\ell]}, \qquad \mathcal{U}_H^* := \min_{1 \leq L \leq M} \mathcal{U}_H(L). \tag{G.11}$$

Every combined success value lies in $[0, v_j^{\max}]$ by Envy-Free. If an agent has no early role, its gross utility is bounded by its exact tail term; otherwise it is bounded by H_i. Up-front payments are nonnegative. Hence

$$\sum_{i \in H} u_i \leq \mathcal{U}_H^*. \tag{G.12}$$

Substitution in (M1c) proves

$$R \geq \frac{d}{K} \max\left\{ [c_\delta d A_x(B, H) - \mathcal{U}_H^*]_+, \Psi_\delta(A_r(B, H), \mathcal{U}_H^*) \right\}. \tag{M2}$$

G.3 Probability-competition consequences

Assume every box B has a set H_B of $K \geq 3$ agents satisfying $p_i^B \geq \alpha p_B^{\max}$. Then

$$A_x(B, H_B) \geq \alpha(K-2) z(p_B^{\max}, v_B^{(2)}), \qquad A_r(B, H_B) \geq \alpha(K-2) p_B^{\max} v_B^{(2)}. \tag{G.13}$$

Define

$$H^{\max} := \max_i H_i, \qquad \rho_L := \begin{cases} \delta^L, & L < M, \\ 0, & L = M, \end{cases} \qquad \eta_{K,L} := \rho_L + (1-\rho_L)\frac{\min\{2L, K\}}{K}. \tag{G.14}$$

The tail terms in (G.11) give

$$\mathcal{U}^*_{H_B} \leq K \eta_{K,L} H^{\max}. \tag{G.15}$$

Also

$$\max_B z(p_B^{\max}, v_B^{(2)}) \geq W^{(2)}, \qquad \max_B p_B^{\max} v_B^{(2)} \geq \frac{W^{(2)}}{G_M(\delta)}. \tag{G.16}$$

Apply (M2) to every target and use (G.13)–(G.16). This gives the common bound

$$R \geq d \max_{1 \leq L \leq M} \max \left\{ \left[c_\delta d \alpha \left(1 - \frac{2}{K}\right) W^{(2)} - \eta_{K,L} H^{\max} \right]_+, \right.$$
$$\left. \Psi_\delta \left(\frac{\alpha}{G_M(\delta)} \left(1 - \frac{2}{K}\right) W^{(2)}, \eta_{K,L} H^{\max} \right) \right\}. \tag{M3}$$

Equations (M1)–(M3) prove the ordinary mixed-equilibrium revenue bounds. All deviations were chosen independently of the realized opponents' reports. No value competition or interior equilibrium-report assumption entered the proof.

Appendix H: Constant Probability Competition with Reported Values

Theorem 5 (Reported-values comparison). *For every $0 < \alpha, \beta \leq 1$ and $0 < \delta < 1$, there is a finite probability depth $K_0(\alpha, \beta, \delta)$, independent of the number of boxes, such that every existing pure equilibrium with interior probability reports that satisfies constant probability competition, qualified value competition, and pointwise exact writer-allocation Envy-Free satisfies*

$$\mathrm{Rev}_y = \Omega\Big(\alpha\beta(1-\delta)^3 \,\mathrm{OPT}_{\text{RV-DNBR}}\Big). \tag{SV-COMP}$$

Proof. Fix an equilibrium satisfying pointwise exact writer-allocation Envy-Free and constant probability competition. For the comparison with dynamic NBR, also assume a pure equilibrium in the interior probability-report domain of Theorem 4.

Put $p_j^{\max} := \max_i p_i^j$, $v_j^{\max} := \max_i v_i^j$, let $v_j^{(2)}$ be the second-highest true value on box j, and define $W^{\max} := W_\delta(p^{\max}, v^{\max})$ and $W^{(2)} := W_\delta(p^{\max}, v^{(2)})$. Assume qualified value competition $v_j^{(2)} \geq \beta v_j^{\max}$. Benchmark monotonicity and homogeneity give

$$H^{\max} = \max_i W_\delta(p_i, v^{\max}) \leq W^{\max} \leq \beta^{-1} W^{(2)}. \tag{H.1}$$

Define

$$\theta := c_\delta(1-\delta)\alpha\beta, \qquad L_0 := \left\lceil \frac{\log(8/\theta)}{\log(1/\delta)} \right\rceil,$$

$$K_0(\alpha,\beta,\delta) := \max\left\{4, \left\lceil \frac{16L_0}{\theta} \right\rceil \right\}. \tag{H.2}$$

Choose $K = K_0$ and $L = \min\{M, L_0\}$ in the first branch of (M3). Definitions (G.14) and (H.2) give

$$\eta_{K,L} \le \frac{\theta}{4}, \qquad 1 - \frac{2}{K} \ge \frac{1}{2}. \tag{H.3}$$

Substituting (H.1)–(H.3) into (M3) proves

$$\mathrm{Rev}_y \ge \frac{c_\delta \alpha (1-\delta)^2}{4} W^{(2)} \ge \frac{c_\delta \alpha \beta (1-\delta)^2}{4} W^{\max}. \tag{H.4}$$

The same proof applies to a pure equilibrium through its degenerate mixed representation. Thus (H.4) gives the intermediate revenue lower bound in terms of $W^{\max}$ and shows explicitly that K_0 is independent of M.

If the equilibrium is pure and also lies in the interior probability-report domain of Theorem 4, combine (H.4) with (B.2):

$$\mathrm{Rev}_y \ge \frac{c_\delta \alpha \beta (1-\delta)^2}{4 G_M(\delta)} \,\mathrm{OPT}_{\text{RV-DNBR}} \ge \frac{c_\delta \alpha \beta (1-\delta)^3}{4} \,\mathrm{OPT}_{\text{RV-DNBR}}\,. \tag{H.5}$$

The first inequality is the sharper finite-M comparison; the second uses $G_M(\delta) \le (1-\delta)^{-1}$. This proves Theorem 5.

Appendix I: The Repair Potential and the Existence Boundary

We show that, under known box values, coordinatewise report-raising repairs weakly increase RW revenue, strictly increase a repair potential, and converge. Order-changing deviations prevent this convergence argument from establishing pure-equilibrium existence. Let $x = \widehat{q}$ be the vector of adopted reports. Suppose only coordinate j rises from x_j to $x_j' > x_j$, and evaluate the new reports in the old order. If $D_j(x)$ is the old reported discounted reach, then

$$\mathrm{Rev}_{\mathrm{RW}}(x') - \mathrm{Rev}_{\mathrm{RW}}(x) \ge (1-\delta) D_j(x) v_j (x_j' - x_j). \tag{I.1}$$

To prove (I.1), let $W_{j+}(x)$ be the continuation following j. The old-order change is

$$D_j(x)(x_j' - x_j)(v_j - \delta W_{j+}(x)).$$

Index order gives $W_{j+}(x) \le \widehat{z}_j(x) \le v_j$, so the bracket is at least $(1-\delta)v_j$. Reoptimizing can only raise revenue.

Revenue need not rise strictly when $D_j(x) = 0$. For any $\eta_q > 0$, however,

$$\Phi(\widehat{q}) := \mathrm{Rev}_{\mathrm{RW}}(\widehat{q}) + \eta_q \sum_j \widehat{q}_j \tag{I.2}$$

strictly increases under every nontrivial coordinatewise bid-raising repair. Every adopted coordinate is nondecreasing and bounded by one, so every repair sequence converges coordinatewise. RW revenue is the maximum of finitely many continuous fixed-order objectives and therefore converges as well.

The missing implication is from convergence to Nash equilibrium. A profitable order-changing deviation may assign the deviator a negative position on the promoted box while increasing the reach of a pre-existing positive role. No nonnegative box-level supporter need exist for the promoted report, so the proposed support-preservation lemma is false. The strict secondary term in (I.2) supplies no revenue multiplier and does not eliminate simultaneous or order-changing deviations at the limit. Thus (I.1)–(I.2) prove monotonicity, strict repair potential, and convergence, but neither existence nor nonexistence of a pure equilibrium.

Appendix J: Pure-Equilibrium Nonexistence Below the Unit Report Cap

Consider RW under known box values with probability reports restricted to $[0, 1)$. Every adopted report can then be strictly overtaken.

Proposition J.1 (Open-domain nonexistence). *There is a three-box, eight-agent instance with known box values and a constant $\varepsilon_0 > 0$ such that, when $q_i^j \in [0, 1)$, the RW game has no pure ε-equilibrium for any $\varepsilon < \varepsilon_0$, under any complete tie rule. Every true success probability in the instance is at most $3/5$.*

Set

$$\delta = \frac{4}{5}, \qquad (v_A, v_B, v_C) = (100, 1, 1). \tag{J.1}$$

There are four anchor agents and four tail agents, with true success probabilities

	p^A	p^B	p^C
H_1	0.50	0	0
H_2	0.50	0.01	0
H_3	0.50	0	0.01
H_4	0.50	0.01	0.01
T_1	0.20	0.50	0.60
T_2	0.20	0.49	0.59
T_3	0.20	0.48	0.58
T_4	0.20	0.47	0.57.

(J.2)

In particular, $p_i^j \le 3/5$ for every agent and box.

Fix an arbitrary pure report profile and an arbitrary resolution of all its ties. Write $a = \widehat{q}_A$. Because there are only three reader roles, at least one anchor agent and at least one tail agent hold no reader role; each such agent has utility zero.

Suppose first that $a < 0.49$. A role-free anchor reports

$$q'_A = \frac{a + 1/2}{2} > a$$

on A and zero on the tail boxes. Since $q'_A \ge 1/4$, box A is strictly first. The deviation gives utility

$$100\left(\frac{1}{2} - q'_A\right) = 50\left(\frac{1}{2} - a\right) > \frac{1}{2}. \tag{J.3}$$

Any zero-report tail role assigned to the deviator has zero payment and nonnegative gross value, so it cannot reduce this gain.

Now suppose that $a \ge 0.49$. Then A is strictly first, since its reported index exceeds the largest possible index of either unit-value tail box. Let $x = \widehat{q}_B$ and $y = \widehat{q}_C$, and choose a role-free tail agent. Because $x, y < 1$, that agent can choose

$$\max\{x, y\} < r_C < r_B < 1, \tag{J.4}$$

report (r_B, r_C) on (B, C), and strictly induce the order A, B, C. For a tail type $(0.20, p_B, p_C)$, its gross value from the two tail positions converges, as $r_B, r_C \uparrow 1$, to

$$G(p_B, p_C) = 0.8(1 - 0.20)p_B + 0.8^2(1 - 0.20)(1 - p_B)p_C. \tag{J.5}$$

Across the four tail types, the smallest value of G is 0.4554752. The payment on C converges to zero, while the payment on B converges to $0.8(1-a)$. Hence the deviator's utility converges to at least

$$0.4554752 - 0.8(1-a) \geq 0.4554752 - 0.8(0.51) = 0.0474752. \tag{J.6}$$

Thus, for every $\varepsilon < \varepsilon_0 := 0.0474752$, reports satisfying $(J.4)$ can be chosen close enough to one to give a role-free agent a gain greater than ε. The two cases cover every report profile. All relevant takeovers and box-order comparisons are strict, so the argument is independent of the complete tie rule. This proves Proposition J.1.

The proposition does not apply to the maintained closed report domain $[0,1]$. Indeed, the same construction admits a tie-supported pure equilibrium under a suitable report-blind, agent-specific tie rule when reports equal to one are allowed. Universal pure-equilibrium existence on $[0,1]$ therefore remains open.

Remark (bidding language). This failure may point to a limitation of the bidding language. The single report q_i^j simultaneously describes success and determines the box's contribution to reported reach. In the construction, some agents would prefer to express a priority stronger than any admissible probability report—informally, they would like to report $q_i^j > 1$. A richer mechanism might instead ask every agent to report two scalars for each box: its reach probability and its success probability. Decoupling these reports may better represent such preferences and may restore equilibrium in instances of this kind. We leave the design and equilibrium analysis of such a mechanism open.